

David Zimmerman

Senior Data Engineer

davidzim007@outlook.com | (229) 306-4443 | New York, NY 10023

SUMMARY

Senior Healthcare Data Engineer with 10+ years of experience designing secure, enterprise-grade data platforms in regulated healthcare environments. At Deloitte, I have led modernization initiatives migrating legacy clinical and claims ETL systems into scalable Azure-based Lakehouse architectures using **Azure Databricks, ADLS Gen2, and Delta Lake**, enabling governed, analytics-ready datasets across multi-hospital networks.

My expertise includes ingestion and transformation of **HL7 v2/v3, FHIR, CDA, and X12 healthcare data formats**, implementation of HIPAA-aligned access controls, and building secure ELT workflows using **Azure Data Factory and PySpark**. I focus on data privacy, lineage, auditability, and reliability while ensuring scalable data processing for clinical analytics, regulatory reporting, and AI initiatives.

Recognized for bridging clinical stakeholders and engineering teams, I translate compliance requirements into scalable technical architectures that protect PHI while enabling actionable insights.

RELEVANT SKILL SETS

Cloud Platforms: Microsoft Azure, Azure Data Lake Storage (ADLS Gen2), Azure Databricks, Azure Data Factory (ADF), Azure Synapse Analytics, Azure Functions, Azure Key Vault, Azure Monitor, Azure Active Directory

Healthcare Data Standards: HL7 v2/v3, FHIR (Fast Healthcare Interoperability Resources), CDA, X12 837/835 Claims Data, EHR Integration, Clinical Data Normalization

Data Engineering & Processing: Databricks, Apache Spark, Spark SQL, PySpark, Delta Lake, Batch ETL/ELT Pipelines, Data Lakehouse Architecture, Structured Data Transformation

Data Governance & Security: HIPAA Compliance, PHI Protection, Data Masking, Encryption at Rest & In Transit, Role-Based Access Control (RBAC), Audit Logging, Data Lineage, Unity Catalog

Programming & Query Languages: Python, Advanced SQL, Query Optimization, Window Functions, UDF Development

Orchestration & DevOps: Azure Data Factory Pipelines, Apache Airflow, Git, CI/CD Pipelines, Infrastructure as Code (Terraform)

PROFESSIONAL EXPERIENCE

Senior Data Engineer | Deloitte

Jan 2022 – Present | New York, NY

- Faced with fragmented EHR and claims data across multiple hospital systems creating compliance and reporting risks, architected a centralized **Azure-based Lakehouse** using **ADLS Gen2 and Azure Databricks**, reducing reporting latency by 40% while strengthening enterprise-wide **data governance** controls.
- To support regulatory interoperability mandates, designed scalable ingestion pipelines parsing **HL7 v2/v3 and FHIR resources** into normalized analytical schemas, significantly improving cross-system compatibility and reducing reconciliation errors across clinical datasets.

- Identified vulnerabilities in **PHI access management**, implemented **Role-Based Access Control (RBAC)**, **encryption at rest and in transit**, and Azure Key Vault integration aligned with **HIPAA compliance**, reducing audit risk exposure.
- As growing clinical volumes caused SLA breaches, optimized distributed **Apache Spark workloads in Databricks**, tuning partitions and cluster configurations to reduce processing time by 33% while preserving data integrity guarantees.
- Addressed ingestion delays impacting executive reporting by developing enterprise-grade **Azure Data Factory (ADF) pipelines** integrating EHR, lab, and claims systems, improving reliability and decreasing manual intervention by over 35%.
- Recognizing the need for regulatory traceability, implemented end-to-end **data lineage tracking and audit logging** using Azure Monitor, ensuring full visibility into PHI data movement for compliance reporting.
- Standardized **Bronze/Silver/Gold Lakehouse layers** leveraging **Delta Lake ACID transactions**, improving data quality enforcement and simplifying downstream BI and regulatory analytics workflows.
- Collaborated with clinical SMEs to redesign data models supporting **Medicare and Medicaid reporting**, enabling more accurate reimbursement analytics and improved provider performance tracking.
- Reduced Azure infrastructure costs by implementing cluster right-sizing and workload scheduling optimization within **Databricks**, achieving measurable cost savings while maintaining SLA adherence.
- Mitigated recurring data quality issues by implementing automated validation frameworks using **Python and advanced SQL**, detecting schema drift and anomalies before impacting downstream regulatory reporting.
- Led modernization of legacy SQL Server ETL processes into scalable **Azure ELT frameworks**, improving resilience and reducing operational dependency on manual batch processes.
- Mentored engineers on **healthcare data standards, HIPAA-aligned engineering practices, and Spark optimization techniques**, strengthening secure delivery across regulated client environments.

Data Engineer | Inxoft

Aug 2018 – Dec 2021 | Philadelphia, PA

- Faced with limited scalability in on-prem healthcare reporting systems, led migration to **Azure Data Lake and Azure Synapse Analytics**, enabling elastic storage and improving analytics throughput for large-scale clinical datasets.
- Designed dimensional models supporting complex insurance claims and provider performance reporting, aligning technical structures with healthcare compliance requirements.
- Addressed synchronization challenges between transactional EHR systems and reporting warehouses by implementing **CDC ingestion workflows**, reducing data freshness gaps and improving decision-making reliability.
- Developed validation scripts ensuring accurate mapping of **HL7 segments** into structured analytical tables, reducing reconciliation discrepancies across healthcare data sources.
- Optimized warehouse performance by refining advanced **SQL queries and indexing strategies within Azure environments**, improving query execution time by 30% during peak reporting cycles.
- Strengthened enterprise security posture by implementing **encryption standards, RBAC policies, and PHI protection mechanisms**, aligning systems with **HIPAA compliance frameworks**.
- Automated workflow orchestration using **Azure Data Factory and Apache Airflow**, reducing manual scheduling errors and improving SLA adherence across reporting jobs.

- Supported regulatory audit preparation by documenting **data lineage, transformation logic, and compliance workflows**, ensuring readiness for governance reviews.
- Improved pipeline observability by implementing **Azure Monitor logging and alerting frameworks**, reducing incident response time and increasing operational transparency.
- Contributed to enterprise cloud architecture discussions defining scalable, secure data engineering standards for healthcare-focused clients.
- Provided peer mentorship and code reviews reinforcing **secure coding practices and governance-first engineering principles**.
- Partnered with analytics and finance teams to standardize healthcare KPIs supporting executive dashboards and regulatory reporting initiatives.

Python Developer | Fingent

Jun 2015 – Jul 2018 | White Plains, NY

- Faced with inconsistent clinical and financial reporting workflows across hospital systems, designed structured **SQL-based ETL pipelines** that standardized healthcare datasets, improving reporting accuracy and reducing reconciliation delays for operational stakeholders.
- As provider and billing data volumes increased, implemented scalable ingestion workflows integrating transactional databases into centralized reporting environments, strengthening data consistency and improving dashboard latency for healthcare analytics teams.
- Addressed discrepancies in patient encounter and claims records by developing transformation and validation routines using **SQL** and **Python**, significantly improving trust in downstream regulatory and executive reporting outputs.
- Designed normalized and dimensional warehouse schemas aligned with healthcare reporting requirements, supporting provider performance analytics and improving visibility into key clinical and financial KPIs.
- Implemented **Change Data Capture (CDC)** mechanisms to synchronize hospital transactional systems with reporting layers, reducing data freshness gaps and ensuring more reliable availability of regulated datasets.
- Strengthened data reliability and security posture by introducing structured logging, error-handling frameworks, and foundational **access control policies**, contributing to improved protection of sensitive healthcare information.

EDUCATION

Master's Degree in Computer Science, *Columbia University*

May 2015 | New York, NY

Bachelor's Degree in Software Engineering, *Columbia University*

Apr 2013 | New York, NY